

Climate & Agriculture Report

Prepared for: **Varun Family Farm**

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Farm Profile

Location

Armoor, NAGARKURNOOL

Soil Type

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Total Area

- Hectares

Primary Irrigation Source

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Long-term Crop Suitability Projection

Crop	Current Status	Projected (SSP2-4.5)	Change Trend	Recommendation
Turmeric	Marginal	Unsuitable	Improving (+1)	Avoid expansion under high heat stress
Maize	Moderate	Unsuitable	Improving (+2)	Limit to irrigated or low-risk blocks
Paddy	Suitable	Unsuitable	Improving (+3)	Shift area to short-duration paddy or millets
Cotton	Highly Suitable	Unsuitable	Improving (+4)	High risk under SSP585
Millets	Highly Suitable	Unsuitable	Improving (+4)	Prefer hardy short-duration varieties

2030 Climate Projections (SSP2-4.5)

Year	Avg Temp Change	Rainfall Change	Agri Stress Index	Drought Hazard
2030	+0.25°C	+1.29%		2.62

Year	Avg Temp Change	Rainfall Change	Agri Stress Index	Drought Hazard
2040	+0.75°C	+3.79%		8.66
2035	+0.50°C	+2.54%		2.55

Strategic Interventions & Recommendations

Based on the climate modeling for your region, the following structural adjustments are recommended to mitigate long-term climate risks:

- **Irrigation Infrastructure:** Given potential water deficits in critical growth phases, investment in drip or micro-sprinkler systems is highly advised to optimize water use efficiency by up to 40%.
- **Crop Transition:** If Paddy suitability declines below 50% in future SSP scenarios, consider phasing into drought-resistant cash crops such as Millets or specialized Pulses.
- **Soil Moisture Retention:** Increase organic carbon content through mulching and cover cropping to improve the soil's natural water retention capacity during dry spells.